# 2. TJEDAN:

## ~ Svojstva signala ~

**Zadatak 1.) Koji od zadanih su periodički? Temeljni period?**

a) $\cos^2(t)$ uvjet periodičnosti: $x(t) = x(t+T)$

$\cos^2(t) \Rightarrow \cos^2(t+T)$

$\cos^2(t+T) = \frac{\cos 2(t+T)+1}{2}$

$= \frac{\cos 2(t+T)}{2} + \underbrace{\frac{1}{2}}_{\text{konst}}$

$x(t+T) = \frac{1}{2}\cos 2(t+T)$

$= \frac{1}{2}\cos(2t+2T) = \frac{1}{2}\left[\cos 2t \cos 2T - \sin 2t \cdot \sin 2T\right]$

$\cos 2t = \cos^2(t) - \sin^2(t)$

$\cos 2t = \cos^2(t) - 1 + \cos^2(t)$

$\cos^2(t) = \frac{\cos 2t + 1}{2}$

$\cos^2(t) = \frac{1}{2} + \frac{\cos 2t}{2}$

konst

$x(t) = \frac{1}{2}\cos(2t)$

ako vrijedi periodičnost $x(t+T) = x(t)$ onda mora vrijediti da je $T = 2k\pi$

jer je $\cos(x) = \cos(x+2\pi)$

$\Rightarrow \sin 2t \cdot \sin 2T = \emptyset \longrightarrow x(t+T) = \frac{1}{2}\cos 2t \cdot \underbrace{\cos 2T}_{1}$

$$\boxed{x(t+T) = \frac{1}{2}\cos 2t}$$

Period:

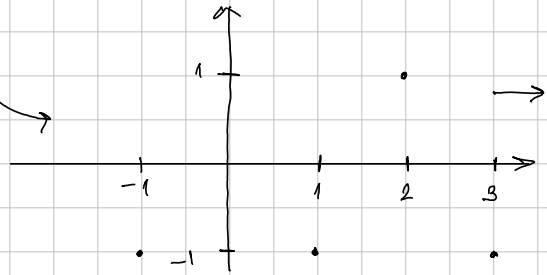
$\cos(2T) = 1 \,/\arccos \Rightarrow \cancel{2}T = \cancel{2}k\pi$ osnovni period $\Rightarrow k = 1$

$$\boxed{T = \pi}$$

b) $\cos(2\pi t)\cdot u(t) \rightarrow$ nije periodičan čim je pomnožen sa stepom

c) $e^{j\pi t} = \cos \pi t + \underbrace{j\sin \pi t}_{0}$

$\cos \pi t$




$\rightarrow$ periodičan

cos je periodičan za $2k\pi$

$\boxed{T = 2\pi}$

d) $\cos(t^2)$ $\quad x(t) = \cos(t^2) \longrightarrow x(t) = x(t+T)$

$x(t+T) = \cos((t+T)^2) \Rightarrow \cos(t^2 + 2tT + T^2) = \cos(t^2)$ / arccos

$\cancel{t^2} + 2tT + T^2 = \cancel{t^2} \longrightarrow T^2 + 2t\cdot T = 2k\pi$

$T^2 + 2t\cdot T - 2k\pi = 0$

$$T_{1,2} = \frac{-2t \pm \sqrt{4(2t)^2 + 8k\pi}}{2} = \frac{-2t \pm \sqrt{16t^2 + 8k\pi}}{2} = \frac{-2t \pm 2\sqrt{4t^2 + 2k\pi}}{2}$$

$$\boxed{T_{1,2} = -t \pm \sqrt{\underbrace{4t^2 + 2k\pi}_{\geq 0}} \rightarrow T \in \mathbb{R}}$$

e) $e^{j\omega_0 t} = \cos \omega_0 t + j\sin \omega_0 t$

$x(t+T) = e^{j\omega_0 (t+T)} = e^{j\omega_0 t} \cdot \underbrace{e^{j\omega_0 T}}_{\text{konst}} \longrightarrow x(t) = x(t+T)$ ✓

$e^{j\omega_0 T} = 1$

$\cos \omega_0 T = 1 \qquad \omega_0 T = 2k\pi \longrightarrow \boxed{T = \frac{2\pi}{\omega_0}}$

Zadatak 2.) Jesu li periodički? $T_0 = ?$

a) $\cos\left(\pi n + \frac{\pi}{4}\right)$

uvjet periodičnosti za diskretne signale: $x[n] = x[n+N]$

period $= N$

$$x[n] = \cos\left[\pi n + \frac{\pi}{4}\right]$$

$$x[n+N] = \cos\left[\pi(n+N) + \frac{\pi}{4}\right] = \cos\left[\pi n + \pi N + \frac{\pi}{4}\right]$$

$$\cancel{x[n] = x[n+N] \rightarrow \cos\left[\pi n + \frac{\pi}{4}\right] = \cos\left[\pi n + \pi N + \frac{\pi}{4}\right]}$$

$$\cancel{\pi n + \frac{\pi}{4} = \pi n + \pi N + \frac{\pi}{4} \rightarrow \pi N = 2k\pi}$$

$$\cos\left[\pi n + \pi N + \frac{\pi}{4}\right]$$

$$\cancel{N = 2k \longrightarrow \boxed{N_0 = 2}}$$

$$\cos\left[\pi n + \frac{\pi}{4}\right]\cos[\pi N] - \cancel{\sin\left[\pi n + \frac{\pi}{4}\right]\cdot \underbrace{\sin[\pi N]}_{0}} \qquad N \in \mathbb{N}$$

$$\Rightarrow \cancel{\cos\left[\pi n + \frac{\pi}{4}\right]}\cos[\pi N] = \cancel{\cos\left[\pi n + \frac{\pi}{4}\right]}$$

$$\cos[\pi N] = 1 \implies \cancel{\pi} N = 2k\cancel{\pi}$$

$$N = 2k \longrightarrow \boxed{N_0 = 2}$$

b) $\cos\left(\frac{\pi n^2}{8}\right) = x[n]$

$$x[n+N] = \cos\left(\frac{\pi(n+N)^2}{8}\right) = \cos\left(\frac{\pi}{8}(n^2 + 2nN + N^2)\right) = \cos\left(\frac{\pi}{8}n^2 + \frac{\pi}{8}(2nN + N^2)\right)$$

$$\frac{\pi}{8}(2nN + N^2) = \frac{2\pi n N}{8} + \frac{\pi}{8}N^2 \xrightarrow[\text{periodično}]{\text{da bi bilo}} \frac{2\pi n N + N^2\pi}{8} = 2k\pi$$

$$\cancel{N^2 \neq 2\pi n N - 16k\pi = 0}$$

$$\pi \frac{N}{8}\cdot(2n + N) = 2k\pi \quad /:2\pi$$

$$\frac{N}{8}\left(n + \frac{N}{2}\right) = k$$

$n, N \in \mathbb{N}$, $k \in \mathbb{Z}$ $\rightarrow$ ne smije biti razlomak $\rightarrow$ $N = 8\ell$ (višekratnik)

$$\boxed{\text{temeljni period} \Rightarrow N = 8}$$

c) $\cos\left(\frac{n}{3}+1\right) = x[n]$

$$x[n+N] = \cos\left(\frac{n+N}{3}+1\right) = \cos\left(\frac{n}{3}+\frac{N}{3}+1\right)$$

$$\frac{N}{3} = 2k\pi$$

$$\boxed{N_0 = 6\pi} \quad \lightning \quad N \in \mathbb{N}!$$

$\pi \notin \mathbb{N} \rightarrow$ nije periodičan

Zadatak 3.) $x(t) = e^{j\omega_0 t}$

kontinuirani kompleksni eksp. signal.

$x[n] \leftarrow x(t)$ dobiven uniformnim očitavanjem s periodom $T_s$.

↳ je li periodičan?

$$x(t) = e^{j\omega_0 t} = \cos\omega_0 t + j\sin\omega_0 t$$

$$\downarrow$$

$$t = T_s N$$

$$\downarrow$$

$$x[n] = x[n\cdot T_s] = e^{j\omega_0 n T_s}$$

$$x[n+N] = e^{j\omega_0 T_s(n+N)} = e^{j\omega_0 T_s \cdot n} \cdot e^{j\omega_0 T_s \cdot N}$$

$$x[n] = x[n+N] \longrightarrow e^{j\omega_0 n T_s} = e^{j\omega_0 n T_s} \cdot e^{j\omega_0 N T_s}$$

$$e^{j\omega_0 N T_s} = 1$$

$$\omega_0 N T_s = 2k\pi \rightarrow N = \frac{2k\pi}{\omega_0 \cdot T_s} \quad k \in \mathbb{Z}, \quad N = \frac{2\pi}{\Omega_0} k, \in \mathbb{N}$$

$\Omega = \frac{m}{\ell}\pi$ do kraja skraćen $\frac{m}{\ell}$

$m \in \mathbb{Z}, \ell \in \mathbb{N}$

$$\Rightarrow N = \frac{2m}{\ell} k \quad \begin{cases} 2\ell, & \text{za } m \text{ neparan } (k=m) \\ \ell, & \text{za } m \text{ paran } \left(k = \frac{m}{2}\right) \end{cases}$$

$$N \in \mathbb{N}, \quad T_0 = \frac{2\pi}{\omega_0} \rightarrow \omega_0 = \frac{2\pi}{T_0} \Longrightarrow \boxed{N = \frac{2k\pi}{\frac{2\pi}{T_0}\cdot T_s} = \frac{T_0}{T_s} k}$$

→ budući da $N \in \mathbb{N}$, $\frac{T_0}{T_s}$ mora dati $\mathbb{N}$ $(k \in \mathbb{Z})$

Zadatak 4.) $x[n] = \cos(an+1)$. Kakav mora biti a da bi signal bio periodičan?

$$x[n] = \cos(an+1)$$

$$x[n+N] = \cos(an+1+aN) \rightarrow aN = 2k\pi$$

$$N = \frac{2k\pi}{a} = \frac{2l}{m}k$$

$$\frac{\pi}{a} = \frac{l}{m}$$

$$a = \pi\frac{m}{l} \; ; \; l, m \in \mathbb{Z}$$

$$\boxed{N = \frac{2k\cancel{\pi}}{\cancel{\pi}\cdot m}\cdot l = \frac{2kl}{m}}$$

budući da su $l, m, k \in \mathbb{Z} \rightarrow N \in \mathbb{N}$ ✓

Zadatak 5.) $g: \mathbb{Z} \rightarrow \mathbb{R}$, $f: \mathbb{Z} \rightarrow \mathbb{R}$

$$\forall n \in \mathbb{Z}, \; f[n] = \sum_{k=-\infty}^{\infty} g(n-kp), \; p \in \mathbb{N}$$

Dokažite da je signal f periodičan za svaki diskretan signal g za kojeg suma konvergira.

$$f[n+N] = \sum_{k=-\infty}^{\infty} g(n+N-kp) \qquad m, p \in \mathbb{N} \rightarrow N = m\cdot p$$

$$f[n+N] = \sum_{k=-\infty}^{\infty} g(n-p(k-m))$$

Želimo takav oblik $\rightarrow t = k-m$

$k = -\infty \rightarrow t = -\infty$

$k = \infty \rightarrow t = +\infty$

$$\boxed{f[n+N] = \sum_{t=-\infty}^{\infty} g(n-pt) = f[n] \; ✓}$$

periodičan je

## Zadatak 6.)

a) $\int_0^\infty \delta(t-2)\, t^2\, dt =$ * svojstvo Diraca: $f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$

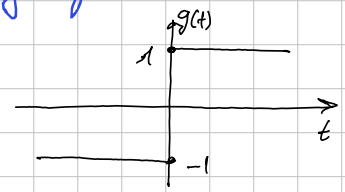
$\int_{-\varepsilon}^{\varepsilon} \delta(t)\, dt = 1$

$= \int_0^\infty \delta(t-2)\cdot 4\, dt = \boxed{4}$

b) $\int_{-\infty}^{\infty} \underbrace{\mu(t-1)\,\delta(t)\,\cos t}_{f(t)}\, dt = \underbrace{\mu(t_0-1)}_{\mu(-1)}\cdot \underbrace{\cos(t_0)}_{1} \underbrace{\int_{-\infty}^{\infty}\delta(t)\,dt}_{1} = \mu(-1) = \boxed{\emptyset}$ $\quad t_0 = 0$

## Zadatak 7.) Provedite i skicirajte generaliziranu derivaciju:

$$g(t) = \operatorname{sgn}(t) = \begin{cases} 1, & t>0 \\ -1, & t<0 \end{cases}$$




$g(t) = -1 + 2\mu(t)$ $\quad \boxed{g'(t) = 0 + 2\delta(t)}$

## Zadatak 8.) Izračunajte generaliziranu derivaciju signala:

a) $g(t) = t(\mu(t) - \mu(t-1)) + (3-t)(\mu(t-2) - \mu(t-3))$

$g(t) = t\mu(t) - t\mu(t-1) + 3\mu(t-2) - 3\mu(t-3) - t\mu(t-2) + t\mu(t-3)$

$g'(t) = \mu(t) + t\delta(t) - \mu(t-1) - t\delta(t-1) + 3\delta(t-2) - 3\delta(t-3) - \mu(t-2)$
$\quad - t\delta(t-2) + \mu(t-3) + t\delta(t-3)$

$g'(t) = [\mu(t) - \mu(t-1)] - [\mu(t-1) - \mu(t-3)]$
$\quad + t[\delta(t) - \delta(t-1)] + 3[\delta(t-2) - \delta(t-3)] - t[\delta(t-2) - \delta(t-3)]$

$g'(t) = [\mu(t) - \mu(t-1)] - [\mu(t-2) - \mu(t-3)] + t[\delta(t) - \delta(t-1)] + (3-t)[\delta(t-2) - \delta(t-3)]$

$\underset{0}{\cancel{t\delta(t)}} \underset{1}{-t\delta(t-1)} + 3\delta(t-2) - 3\delta(t-3)$
$\quad \underset{2}{-t\delta(t-2)} + \underset{3}{t\delta(t-3)}$

$g'(t) = [\mu(t) - \mu(t-1)] - [\mu(t-2) - \mu(t-3)] + \delta(t-1) + 3\delta(t-2) \cancel{-3\delta(t-3)}$
$\quad -2\delta(t-2) \cancel{+3\delta(t-3)}$

$$\boxed{g'(t) = [\mu(t) - \mu(t-1)] - [\mu(t-2) - \mu(t-3)] + \delta(t-1) + \delta(t-2)}$$

b) $g(t) = (3-t)(\mu(t) - \mu(t-1)) + (3-t)(\mu(t-2) - \mu(t-3))$

$$g(t) = 3\mu(t) - 3\mu(t-1) \underline{-t\mu(t)} + \underline{t\mu(t-1)} + 3\mu(t-2) - 3\mu(t-3) \underline{-t\mu(t-2)} \underline{+t\mu(t-3)}$$

$$g'(t) = 3[\delta(t) - \delta(t-1) + \delta(t-2) - \delta(t-3)]$$

$$-\mu(t) - \underset{\downarrow\, 0}{t}\delta(t) + \mu(t-1) + \underset{\downarrow\, 1}{t}\cdot\delta(t-1) - \mu(t-2) - \overset{\uparrow\, 2}{t}\delta(t-2) + \mu(t-3) + \underset{\downarrow\, 3}{t}\delta(t-3)$$

$$\Rightarrow g'(t) = 3[\delta(t) - \delta(t-1) + \delta(t-2) - \cancel{\delta(t-3)}] - [\mu(t) - \mu(t-1) + \mu(t-2) - \mu(t-3)]$$

$$+ \delta(t-1) - 2\delta(t-2) + \cancel{3\delta(t-3)}$$

$$\boxed{g'(t) = 3\delta(t) - 2\delta(t-1) + \delta(t-2) - [\mu(t) - \mu(t-1) + \mu(t-2) - \mu(t-3)]}$$

Zadatak 9.) $x[n] = \cos\left(\frac{\pi n}{8}\right)$.

Dva različita kontinuirana signala koja otipkavanjem dajeu ovaj diskr. signal. Frekvencija otipkavanja neka je $f_s = 10\,kHz$.

$x[n] = \cos\left(\frac{\pi n}{8}\right)$ $\quad x(t) = \cos(\omega t)$ $\quad t = n\cdot T_s$ $\quad f_s = \frac{1}{T_s} =$

$T_s = 0.1\,\mu s$

$x(nT_s) = \cos(\omega_n T_s)$

→ da bi otipkani signal i zadani diskretni signal bili jednaki $x[n] = x(nT_s)$

$$\cos\left(\frac{\pi n}{8}\right) = \cos(\omega\cdot n\cdot T_s) \quad / \arccos$$

$$\frac{\pi n}{8} = \omega_n T_s \Rightarrow \frac{\pi}{8} = 2\pi\frac{f}{f_s} \rightarrow f = f_s\cdot\frac{1}{16}$$

$$\boxed{f = 625\,Hz} \rightarrow \Omega_0 = \frac{\pi}{8}$$

$$\Omega_0 = 2\pi\frac{f}{f_s}$$

$\omega_1 = 1250\pi$

$x(t) = \cos(1250\pi\cdot t)$

periodičnost

$$\cos\left(\frac{\pi n}{8}\right) \Rightarrow \cos\left(2\pi\frac{f}{f_s}n\right) = \cos\left(2\pi\left(\frac{f + f_s\cdot k}{f_s}\right)\cdot n\right)$$

$$f + f_s\cdot k \xrightarrow{k=0} 2\pi\frac{f}{f_s}n$$

$$\xrightarrow{k=1} 2\pi\left(\frac{f + f_s}{f_s}\right)n \rightarrow f_2 = 625 + 10\,kHz$$

$f_2 = 10\,625\,Hz$

$x_2(t) = \cos(21\,250\pi\cdot t)$